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Studierichting: Informatica

Oefeningenreeks 1B nr. 37.1

$$\begin{aligned}8 + 6i &= (x + yi)^2 \\8 + 6i &= x^2 - y^2 + 2xyi\end{aligned}$$

$$\begin{cases} x^2 - y^2 = 8 \\ 2xy = 6 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 8 \\ x^2(-y^2) = -9 \end{cases}$$

$$\begin{aligned}\lambda^2 - 8\lambda - 9 &= 0 \\ \Delta &= 64 + 36 = 100\end{aligned}$$

$$\begin{aligned}\lambda_1 &= \frac{8 + 10}{2} = 9 \\ \lambda_2 &= \frac{8 - 10}{2} = -1\end{aligned}$$

$$\begin{aligned}x^2 = 9 &\Rightarrow x = 3 \\ y^2 = 1 &\Rightarrow y = 1\end{aligned}$$

$$\begin{array}{ll}\text{Wortels:} & w_1 = 3 + i \\ & w_2 = -3 - i\end{array}$$

References